

ISIT307/MTS9307 - BACKEND WEB PROGRAMMING

SUBJECT REVISION



ASSESSMENT

Assessment Items	Percentage of Final Mark		Due Date
	Marks for the Item	Minimum required for a pass	
Assignment 1 (group work – 2 students)	25	N/A	20/07/2026 (check Moodle site)
Assignment 2 (individual work)	25		FT: 13/08/2026 PT: 18/08/2026 (check Moodle site)
Final Exam (TBA)	50	20	Exam week as per schedule
Total	100	50	The mark must be ≥ 50 to pass the subject

TECHNICAL FAIL

- To be eligible for a Pass in this subject a student must achieve a mark of at least 40% in the Final exam
 - Final Exam (20 out of 50)
- Students who fail to achieve this minimum mark & would have otherwise passed may be given a TF (Technical Fail) for this subject.

FINAL EXAM

- 3 Hours

Double-check the personal exam timetable to ensure you are at the right place and time.

FINAL EXAM – QUESTION TYPES

- **Part A: short & long answer (theory) questions**

(4 questions x 5 marks = 20 marks)

- What is ...
- What are ...
- Explain ...
- Compare and contrast
- What are similarities / differences

- **Part B: coding questions**

(5 questions x 6 marks = 30 marks)

- What is the output of the given code fragment & explain the code (1 question)
- Write PHP script (code segment) that will solve the problem ... / given below. (4 questions)

FINAL EXAM - EXAMPLES

- Part A: short & long answer questions (20 marks)
 - Explain the tools that can be used for maintaining web site's state information.
 - Explain the difference between ...

FINAL EXAM - EXAMPLES

- Part B: coding questions (30 marks)
 - What is the purpose and the output produced by each of the following code fragments? Explain your answer and add comments.

```
<?php
$HospitalDepts = array(
    "Anesthesia",
    "Molecular Biology",
    "Neurology",
    "Pediatrics");
array_pop($HospitalDepts);
array_push($HospitalDepts, "Psychiatry", "Pulmonary
Diseases");
?>
```

FINAL EXAM - EXAMPLES

- Part B: coding questions (30 marks)
 - Write a PHP script that will define a class “My_Class” with data member *name*, constructor, get and set accessor functions for the data member, and method for printing the value of the data member. Create an object “my_name” and display the value of the created object. (Object “my_name” can have a value "Jasper").
Create class ‘New_Class’ that will inherit “My_Class”

SUBJECT OUTLINE

- The subject introduces students to open-source programming languages in web development so that they can inexpensively develop sophisticated web applications.
- Students will become familiar with the integration of programming, databases, web-applications, and structural and object oriented programming.
- The subject aims to integrate the previous knowledge which students have gained through subjects on web technologies, web programming and databases to create real-world web applications.

INTRODUCTION

- PHP syntax and basic structure
- Variables and constants
- Data types
- Operators and expressions

FUNCTIONS AND CONTROL STRUCTURES

- Functions in PHP
- Variable scope
- Control structures
- Loop structures
- `include` and `require` statements

MANIPULATING STRINGS

- String manipulation
- String parsing
- Regular expressions

HANDLING USER INPUT

- Autoglobal variables
- Web forms
- Processing form data
- Create an All-in-One form
- Display dynamic content

FILES AND DIRECTORIES

- File permissions
- Work with directories
- Upload and download files
- Write and Read data to files
- Open and close a file stream
- Manage files and directories

ARRAYS

- Indexed arrays
- Associative arrays
- Manipulating arrays
- Sort, combine arrays
- Multidimensional arrays
- Use arrays in Web forms

PHP: OBJECT-ORIENTED PROGRAMMING

- Object-oriented programming concepts
- Class definition
- Use objects in PHP scripts
- Inheritance, Polymorphism, Abstract classes, Interfaces, Traits
- Object serialization
- Model-view-controller (MVC)

WORKING WITH DATABASES USING PHP

- Connect to MySQL from PHP
- Work with MySQL databases using PHP
- Create, modify, and delete MySQL tables with PHP
- Use PHP to manipulate MySQL records and retrieve database records
- PHP prepared statements

MANAGING STATE INFORMATION

- Learn about state information
- Use hidden form fields to save state information
- Use query strings to save state information
- Use cookies to save state information
- Use sessions to save state information

PHP – XML/JSON, PHP - AJAX

DATA STRUCTURES IN PHP

- Working with XML, PHP XML Parsers
- Working with JSON
- Using AJAX with PHP
- Recursion
- Data structures

MY FINAL ADVICES FOR THE EXAM

- Review all lectures, explained examples, and textbook/referenced sources
- Review all labs
- Review the assignments solutions
- Good luck